AI Department’s Analysis of Data Generated by the Spandan Watch

Date: 20 October 2026

Subject: AI-assisted physical-state and focus-monitoring analysis of Spandan Watch telemetry

1. Executive finding

The AI Department analysed the chronological held-out test data generated by the Spandan Watch. The analysis covered **15,113 telemetry observations from 29 officers** and **1,385 focus-monitoring windows**.

The models identified the following overall distribution:

• **Stress:** 56.2%

• **Physical activity:** 4.2%

• **Normal state:** 39.6%

• **Distracted focus windows:** 23.1%

These figures are model-generated screening results and must be interpreted together with duty context, sensor quality, and supervisory or medical review where appropriate.

2. Models used

The analysis was performed using the repository’s best real-duty-data models, not the synthetic-only or lower-performing comparison models:

Analytical function	Model used
Physical-state prediction: Stress, Physical Activity, or Normal	Best real-duty physical-state model
Focus prediction: Focused or Distracted	Best real-duty focus-monitoring model

Both models were applied to the chronological test set reserved for evaluation.

3. Interpretation of the elevated stress percentage

The elevated stress proportion is attributable to the combined operation of the model and the project’s established labeling criteria. The physical-state model identifies stress principally where elevated heart rate or reduced SpO■ is observed during periods without sufficient movement to explain the cardiovascular load. The test data contains a substantial number of such observations, which increases the proportion classified as stress.

This interpretation is subject to the following qualification: the result is an **AI-derived screening estimate**, not a clinical diagnosis and not a confirmed measurement of psychological stress. Sensor artefacts, missing-sensor imputation, duty-specific exertion, environmental conditions, and the heuristic nature of the labels may affect the estimate. Sustained or repeated alerts should therefore be reviewed against the officer’s actual duty circumstances and, where necessary, assessed by an appropriate medical professional.

4. Officer-level findings

The complete officer-level analysis is provided in `officer_summary.csv`. The six officers with the largest test samples are presented below; all 29 officers are included in the accompanying data file.

Officer	Test rows	Stress %	Activity %	Normal %	Distracted windows %	Observation	Recommendation
PSI Kavita Mistry	2344	56.1	7.6	36.3	34.1	Stress exposure is prominent	Review sustained high-HR stationary periods and add a supervisor check
PI Bharat Rathod	2211	60.6	0.6	38.8	12.0	Stress exposure is prominent	Review sustained high-HR stationary periods and add a supervisor check
ASI Meera Makwana	1650	80.1	1.3	18.6	38.4	Stress exposure is prominent	Review sustained high-HR stationary periods and add a supervisor check
PSI Kiran Mistry	1385	45.1	1.8	53.1	10.5	Stress exposure is prominent	Review sustained high-HR stationary periods and add a supervisor check
PI Nilesh Thakor	1205	46.8	3.7	49.5	13.9	Stress exposure is prominent	Review sustained high-HR stationary periods and add a supervisor check
Constable Dinesh Chaudhary	913	55.8	1.5	42.7	26.7	Stress exposure is prominent	Review sustained high-HR stationary periods and add a supervisor check

5. Observations and recommendations

Stress classifications should be reviewed when they persist across consecutive readings, particularly where heart rate is elevated and movement is low.

A single stress or distraction classification should not by itself trigger medical, disciplinary, or operational action.

Physical-activity classifications should be checked against patrol, traffic, arrest, training, and other duty records because movement patterns vary by assignment.

Repeated distraction classifications should prompt proportionate fatigue, hydration, rest, or supervisory checks.

Future deployments should collect confirmed field annotations and sensor-quality indicators to validate and recalibrate the models before any high-consequence decision-making.

6. Supporting figures

• `physical_state_by_officer.png` — officer-level physical-state comparison.

• `distraction_by_officer.png` — officer-level distracted-window comparison.

• `overall_state_mix.png` — overall physical-state distribution.

7. Conclusion

The Spandan Watch telemetry demonstrates the operational potential of AI-assisted monitoring for identifying patterns of cardiovascular load, physical activity, and possible distraction. The present findings support continued monitored deployment and structured validation. They should be used for early screening and timely review, and not as a substitute for clinical assessment, confirmed incident evidence, or professional judgement.

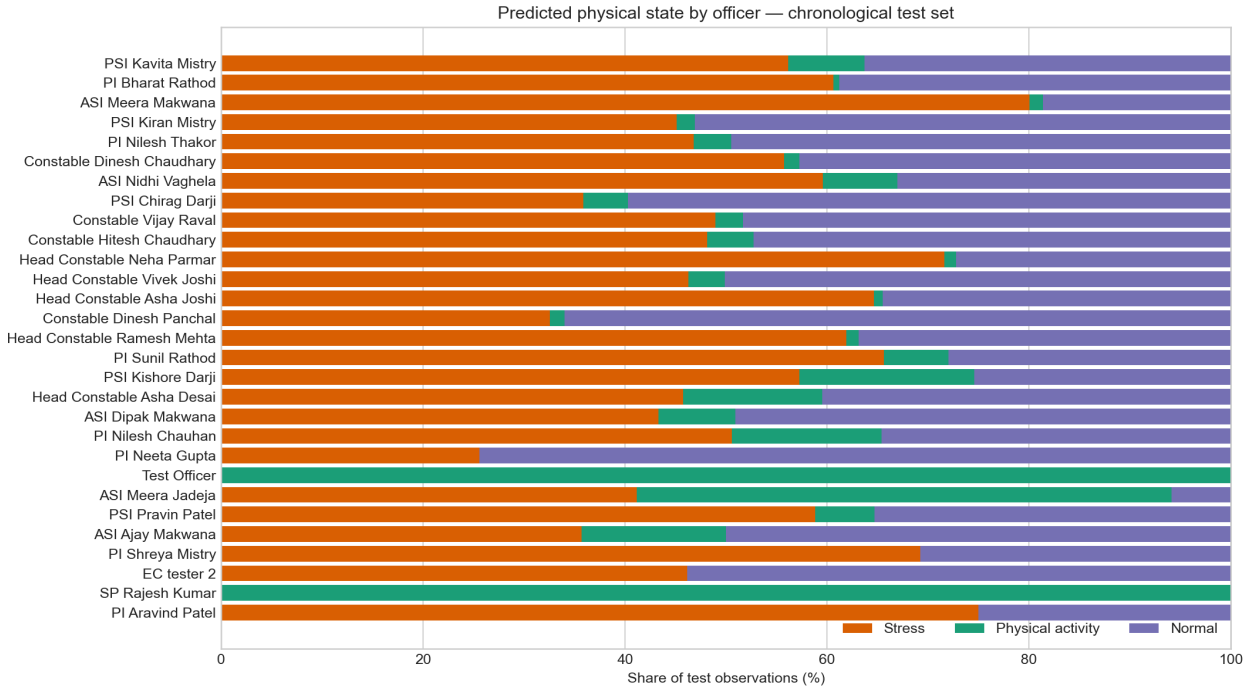


Figure 1. Predicted physical state by officer.

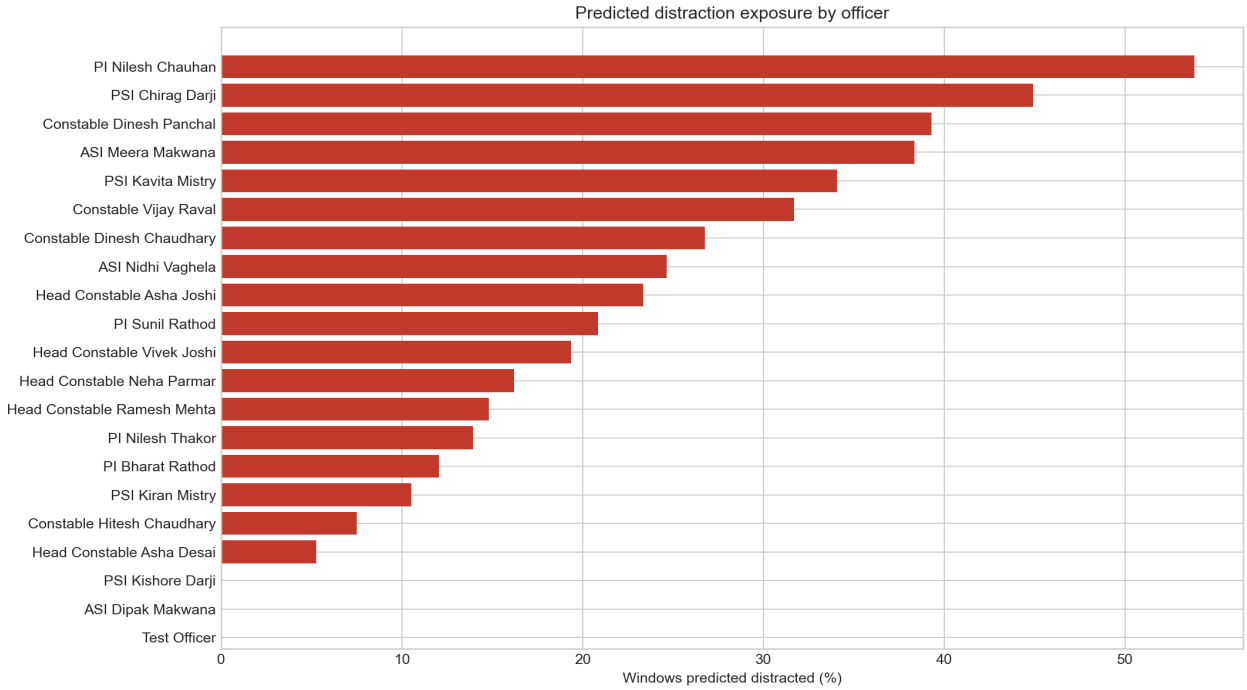


Figure 2. Predicted distraction exposure by officer.

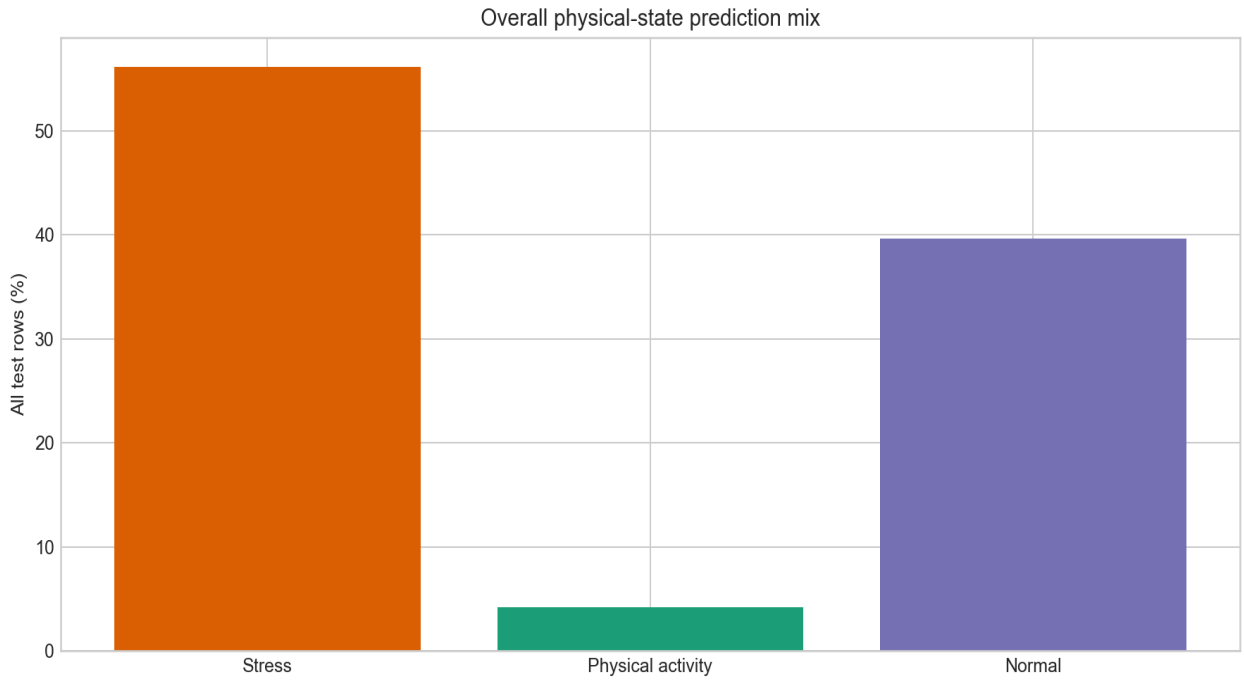


Figure 3. Overall physical-state prediction mix.